EXPERIMENT NO. 03--- BPSK Modulation Scheme



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TITLE: Design of binary phase shift keying (BPSK) modulation scheme

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PROBLEM STATEMENT: Using the PN sequence generated in the previous experiment as data:

1. Design and implement a circuit for the working of modulator and demodulator for the binary phase shift keying (BPSK).
2. Design a circuit showing the working of quadrature phase shift Keying (QPSK).
3. Extract the BPSK modulated data stream, and using MATLAB, demodulate the original data stream (in software domain). Now perturb the BPSK modulated data by adding appropriate noise (additive Gaussian noise) to simulate transmission over an additive Gaussian noise channel, and attempt to demodulate with the same demodulator (take three different noise variance values). Comment on your observation.

KEY COMPONENTS: IC741(Op-Amp), IC74164 (8-bit shift register), ICM4016(Quad Bilateral Switch), IC74LS86, LM710CN, Resistors and capacitors, Potentiometer, Digital oscilloscope, Function generator, Breadboard, Connecting wires, and Power supply.

BRIEF THEORY & BACKGROUND:

The experiment aims to design circuits for the modulation and demodulation of binary data streams by using various digital modulation techniques.

Why Modulation?

In wireless communication, **digital modulation** is the process of superimposing digital information onto a high-frequency carrier for efficient transmission. Compared to analog modulation, it offers higher data capacity, better bandwidth efficiency, improved signal quality, and greater security. By varying a signal's **amplitude (ASK)**, **frequency (FSK)**, or **phase (PSK)**, digital modulation enables reliable communication with reduced noise and distortion. It supports multiplexing of data, voice, and video while maintaining strong resistance to interference. Due to its efficiency, flexibility, low cost, and compatibility with digital systems, digital modulation is the preferred choice for modern wireless communication technologies[1][9].

Advantages of Amplitude Shift Keying (ASK) [2]

- **Simple and cost-effective:** ASK is the easiest digital modulation scheme to implement, requiring minimal circuitry and low-cost components.
- **Low power consumption:** Its simplicity leads to reduced power usage, making it ideal for short-range, low-power applications such as RFID.

- **Suitable for optical communication:** The On-Off Keying (OOK) variant of ASK is widely used in optical fiber communication systems.

Advantages of Frequency Shift Keying (FSK) [2]

- **Robust against noise:** Since information is represented by frequency changes rather than amplitude, FSK is less affected by noise and interference.
- **Simple demodulation:** FSK can be easily implemented and demodulated, especially with non-coherent detection techniques.
- **Constant envelope:** The amplitude remains constant, allowing the use of efficient nonlinear power amplifiers.
- **Reliable for low-speed data:** Commonly used in low-speed applications like caller ID, telemetry, and low-speed modems.

Advantages of Phase Shift Keying (PSK) [2]

- **High bandwidth efficiency:** PSK achieves higher data rates within limited bandwidth, particularly in higher-order schemes like QPSK and 8-PSK.
- **Excellent noise immunity:** Since information is encoded in phase changes, PSK is less affected by amplitude fluctuations.
- **Power-efficient:** Constant envelope PSK schemes permit the use of efficient amplifiers, reducing power requirements.
- **Scalable data rates:** Increasing the number of phase states enables higher data throughput, making PSK suitable for modern systems such as Wi-Fi, 5G, and satellite communications.

What is Binary Phase-Shift Keying (BPSK):

- **BPSK** is a digital modulation technique where the **phase of a carrier signal** is shifted by **180°** to represent binary symbols.
- A **bipolar data stream (±1)** multiplies a **sinusoidal carrier**, producing phase reversals whenever the bit changes.
- The resulting signal has **two distinct phases**: 0° for bit '1' and 180° for bit '0'.
- **Demodulation** is performed by a **synchronous detector** using a carrier of the same frequency and phase.
- The recovered baseband waveform is passed through a **decision circuit** to reconstruct the binary data.
- BPSK offers **good noise immunity** and **power efficiency**, making it widely used in **wireless and satellite communication** systems [4].

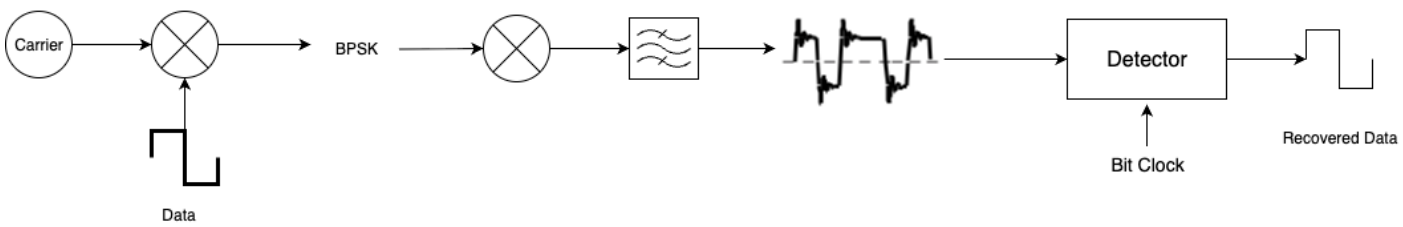


Figure 1: The above is the BPSK Modulation and Demodulation Scheme using Block Diagram [3].

What is Quadrature Phase-Shift Keying (QPSK)?

<Describe the workings of a QPSK scheme>

QPSK Modulation

- **Principle:** This scheme uses two orthogonal basis functions — cosine (in-phase) and sine (quadrature) — for modulation.
- **Symbol-based modulation:** Each **symbol represents 2 bits**, giving **four distinct phases** ($\pi/4$, $3\pi/4$, $5\pi/4$, $7\pi/4$) [9].
- **Constellation:** Has four points on the I–Q plane, representing the in-phase (I) and quadrature (Q) components.
- **Implementation:**
    - A **serial-to-parallel converter** separates the input bit stream into **odd (Q)** and **even (I)** bits.
    - Both streams are converted into **NRZ (±1)** format.
    - The I signal multiplies the **cos($2\pi f_c t$)** carrier; the Q signal multiplies the **sin($2\pi f_c t$)** carrier.
    - The **sum** of these two signals forms the **QPSK modulated waveform** [4].

QPSK Demodulation

- **Technique:** Demodulation uses **coherent detection**, requiring the receiver to know the **carrier frequency and phase**.
- **Carrier synchronization:** Typically achieved using a **Phase-Locked Loop (PLL) Process:**
    - The received QPSK signal is multiplied by **cos($2\pi f_c t$)** and **sin($2\pi f_c t$)** to recover I and Q components.
    - Each output is **integrated over one bit period** to determine the signal polarity.
    - A **threshold detector** decides whether each integrated value corresponds to bit '1' or '0'.
    - The recovered **I (even bits)** and **Q (odd bits)** streams are **recombined** to reconstruct the original data sequence [4].

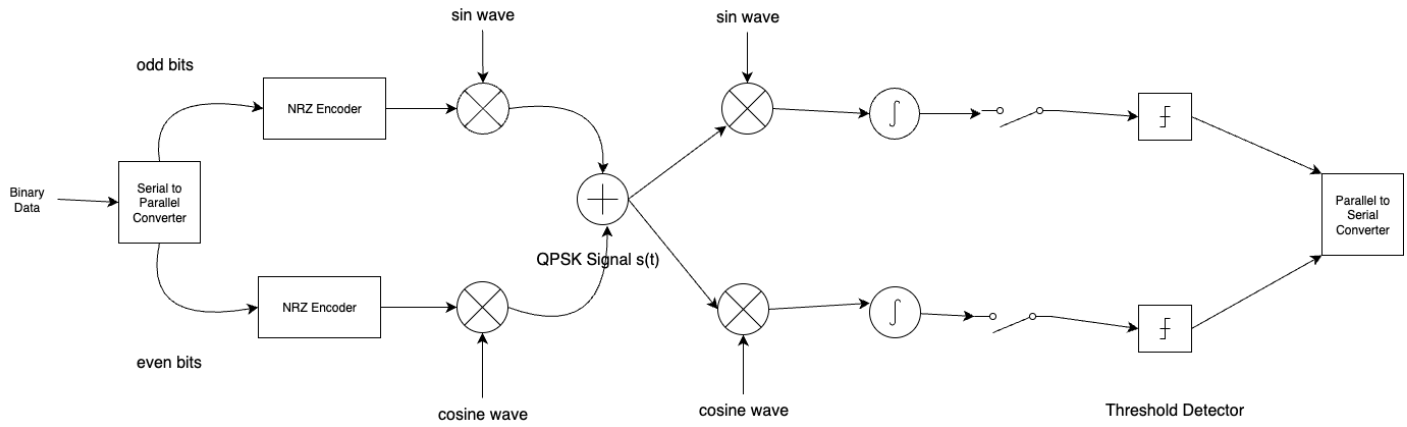


Figure 2: The above is the QPSK Modulation and Demodulation Scheme using Block Diagram [4].

Advantages/Disadvantages of BPSK modulation scheme and their core usage?? Advantages of BPSK

Robustness: BPSK is a robust modulation scheme as the binary 1 and 0 are separated by 180 degree phase shift of the carrier, due to which it can travel longer distances from base stations experiencing less interference [5].

Long-Distance Communication: Due to the robustness, BPSK is typically employed for long distance communications [5].

Power Efficiency: BPSK is a power-efficient modulation technique, as it requires less power to transmit the carrier with a relatively low number of bits [5].

Disadvantages of BPSK

Low Data Rate: Here one bit is carried by one analog carrier, due to which, the data rate in bits per second is the same as the symbol rate [5].

Not Bandwidth Efficient: The data rate is lower compared to other modulation techniques like QPSK (which is twice as efficient) and much lower than higher-order modulation techniques such as 16QAM and 64QAM. Therefore, BPSK is not a bandwidth-efficient modulation technique compared to others [5].

Advantages of QPSK

Increased Bandwidth Efficiency: QPSK carries two bits on a single analog carrier which it twice as bandwidth-efficient compared to BPSK modulation, so the symbol rate is half of the raw bit rate [6].

Bandwidth Efficient: QPSK is a bandwidth efficient modulation technique because a single carrier transmits two bits [6].

Disadvantages of QPSK

Receiver Complexity: In QPSK, each complex symbol represents two binary bits therefore increasing the complexity of the QPSK receiver compared to a BPSK receiver because it needs to distinguish between four states to recover the binary data [6].

Lower Power Efficiency: Due to the increased complexity, QPSK is not as power-efficient as other modulation techniques. More power is required to transmit two bits simultaneously [6].

EXPECTED OUTCOMES:

1. **[Design and circuit implementation]** Design a circuit that modulates and demodulates the previously generated 31-bit PN sequence using the BPSK modulation scheme.



Figure 3 : BPSK Signal Generated for the given PN Sequence

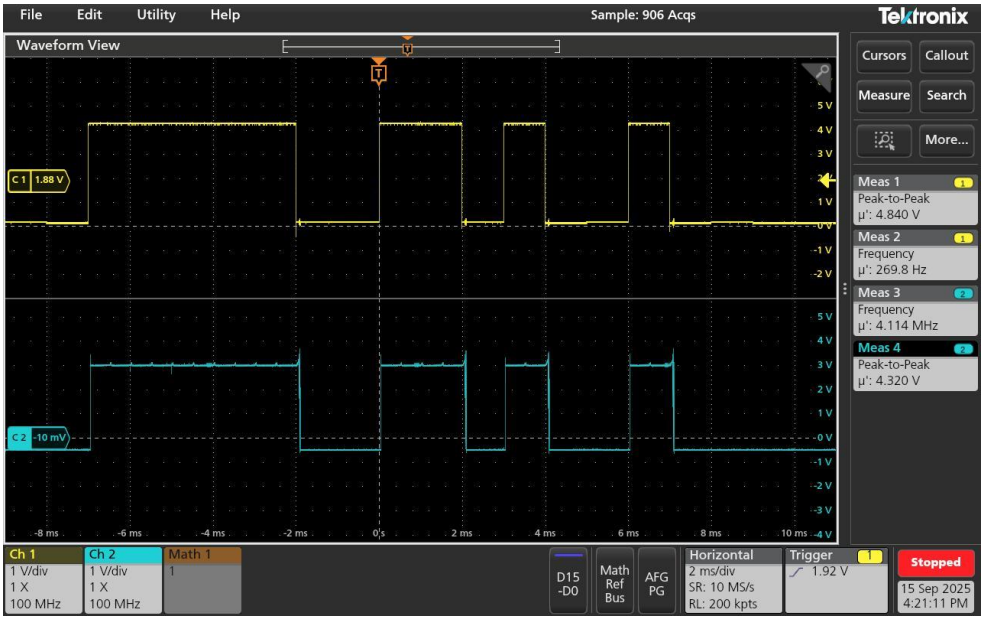


Figure 4 : Demodulated BPSK Signal Generated

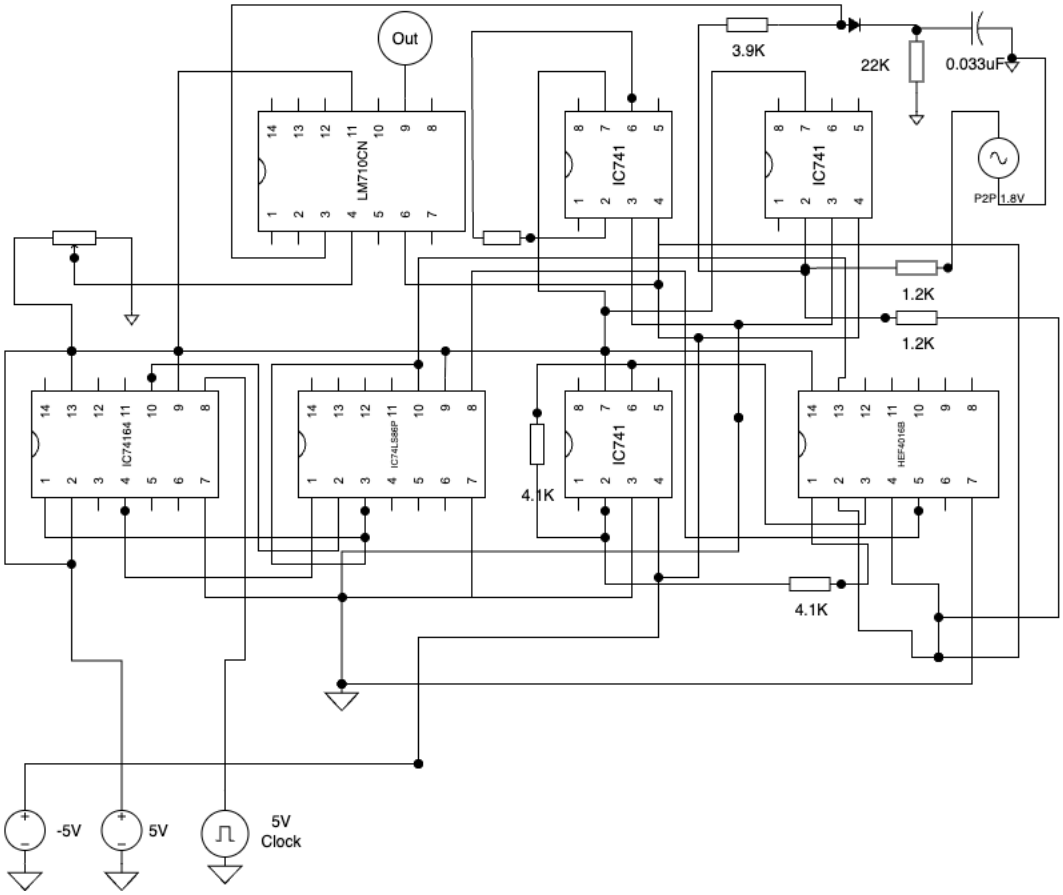


Figure 5: BPSK Modulation and Demodulation Circuit Diagram

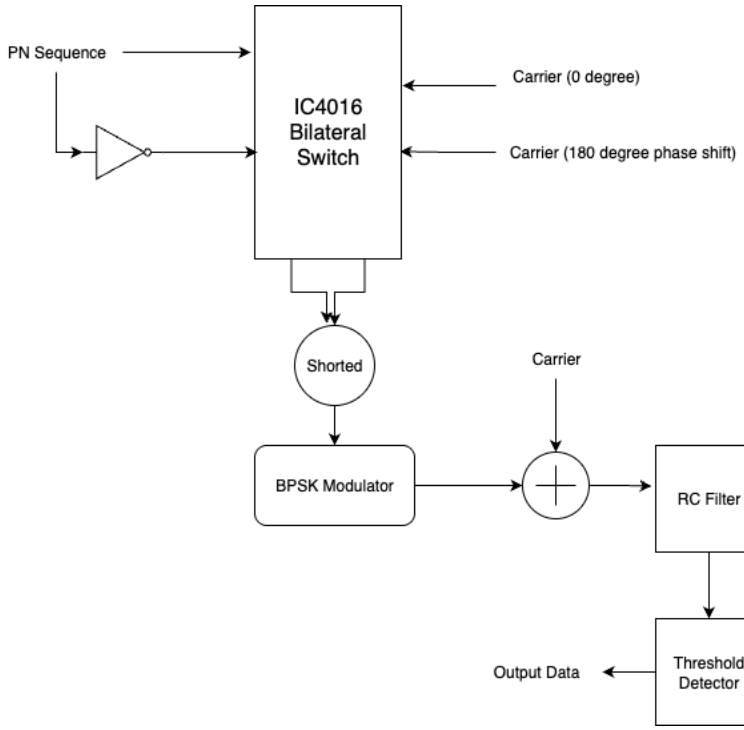


Figure 6: BPSK Block Diagram

2. [Design only] Utilizing the 31-bit PN sequence generated in the previous experiment as data, design a circuit showing the working of Quadrature phase shift Keying(QPSK) [7][8].

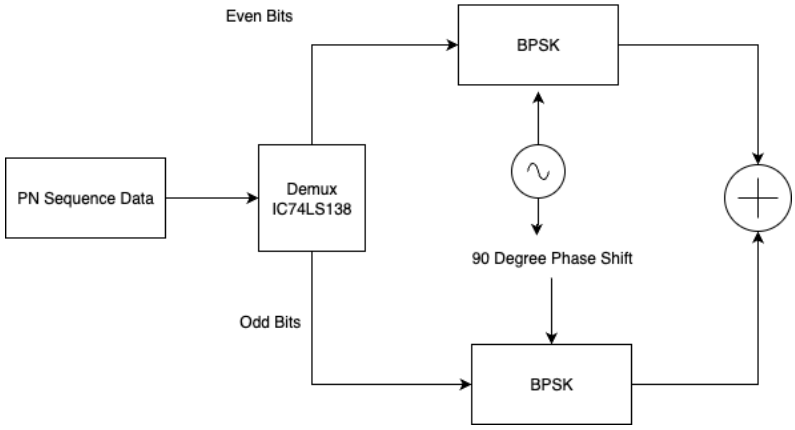


Figure 7 : QPSK Generation using BPSK and PN Sequence [8].

For software parts:

1. Export the BPSK data in .csv format from DSO and build a simple detector. Plot the BPSK modulated and the final demodulated data.

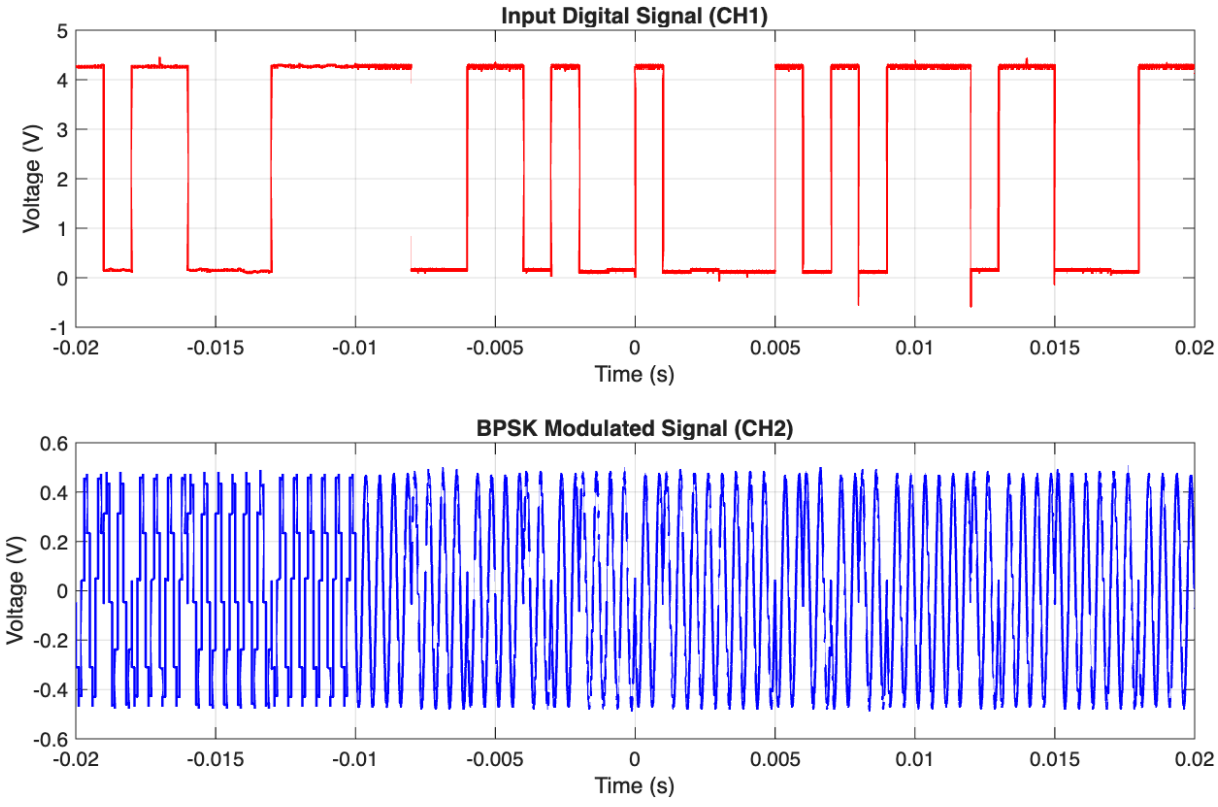


Figure 8 : BPSK Modulated Signal along with the PN Sequence in MATLAB

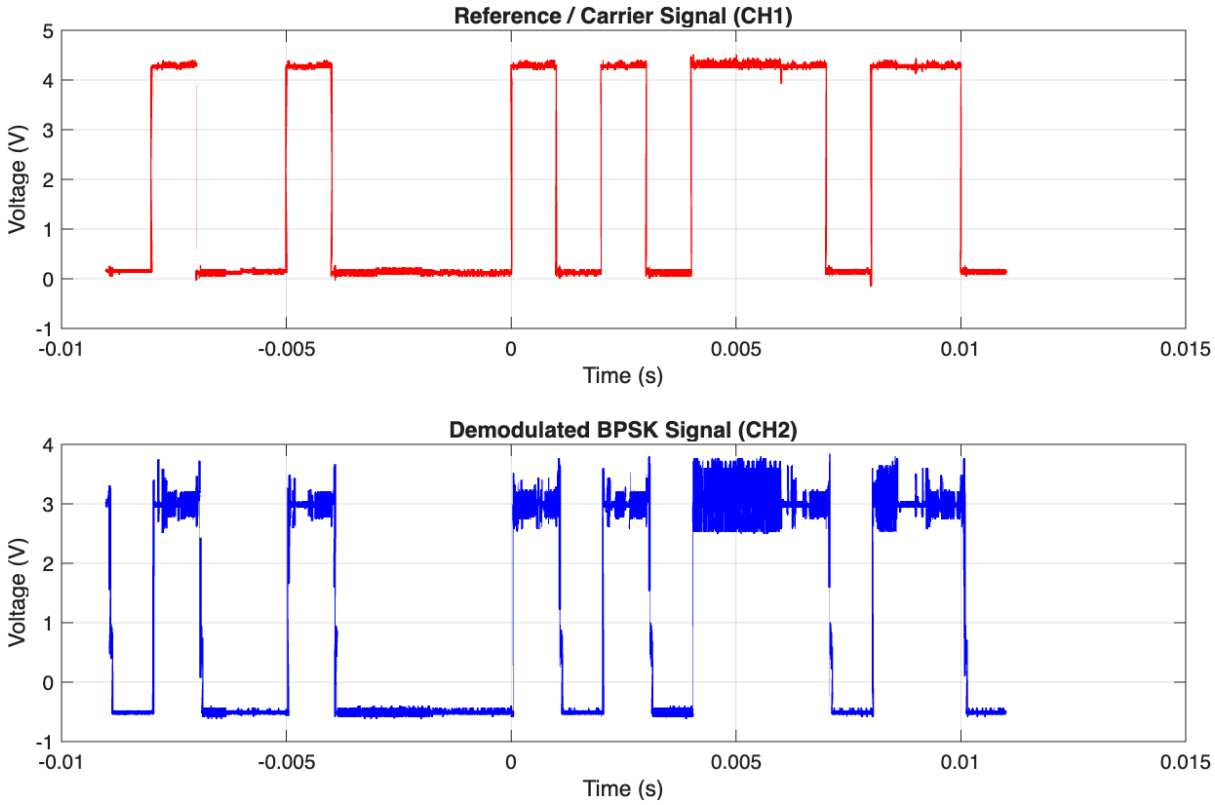


Figure 9 : BPSK Demodulated Signal Along With PN Sequence (Ref Carrier) in MATLAB

2. Export the BPSK data in .csv format from DSO and "simulate" the passing of this data through a noisy channel by adding an independent Gaussian random variable (zero-mean and appropriate variance). Now attempt to decode the original data stream in MATLAB. Repeat for at least three different values of noise variances (low, medium and high) and observe the "error" in decoding.

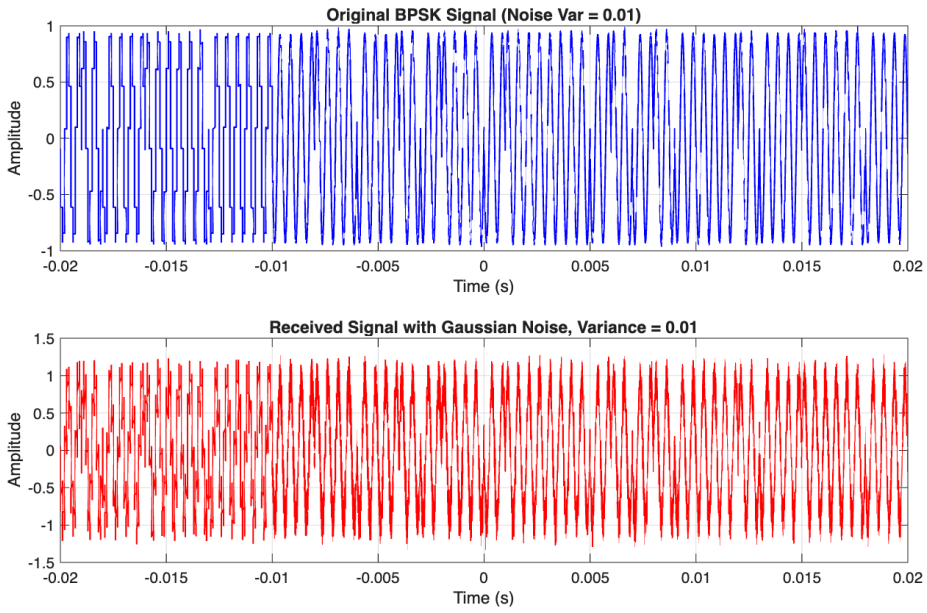


Figure 10: Signal Received after adding gaussian noise with variance 0.01

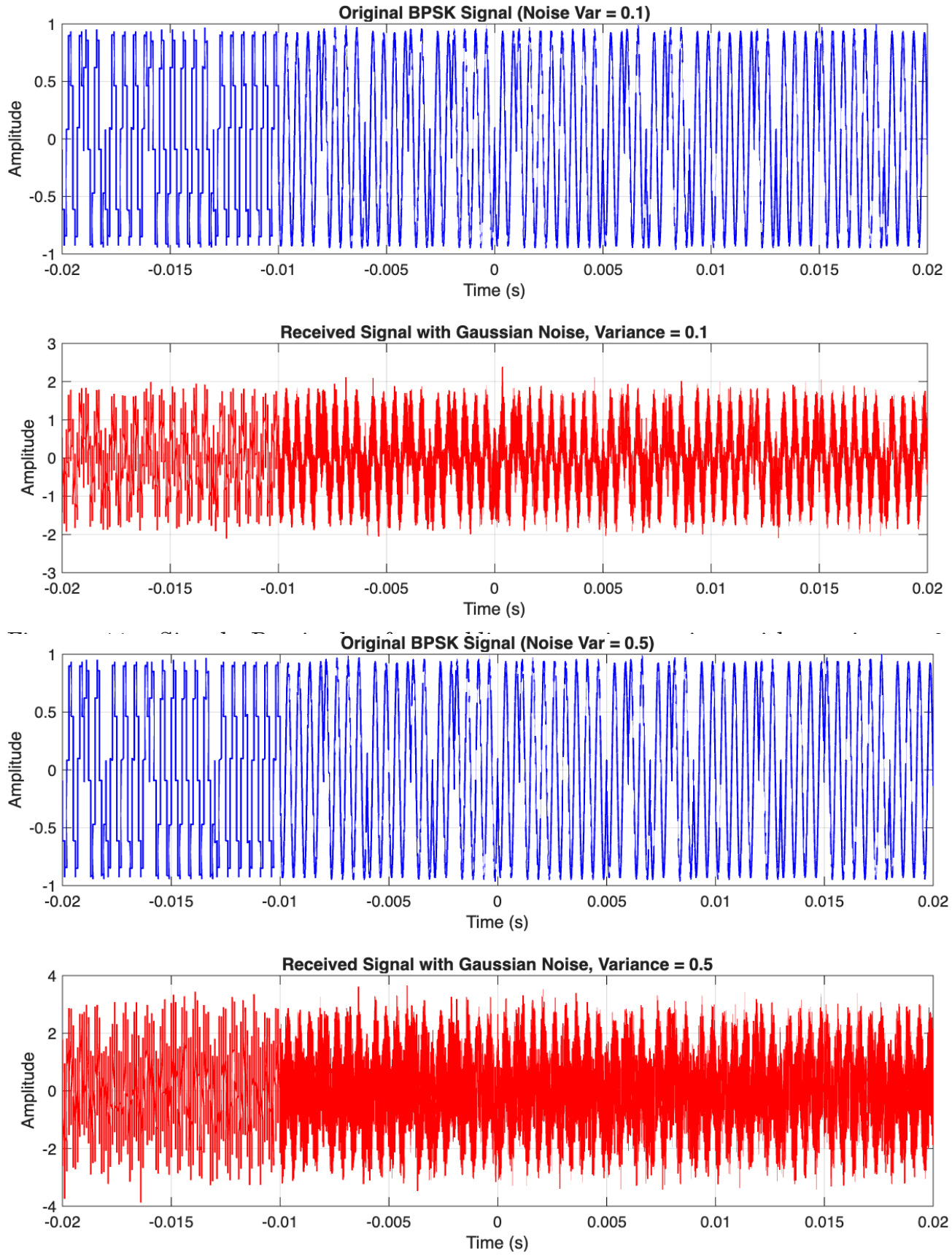


Figure 12 : Signal Received after adding gaussian noise with variance 0.5

3. Construct an average error probability versus noise variance plot. For this plot, run this experiment N=256 times (for each value of noise variance) and plot the average probability of error as a function of the noise variance (as noise variance increases, the signal-to-noise ratio decreases). Comment on your observations.

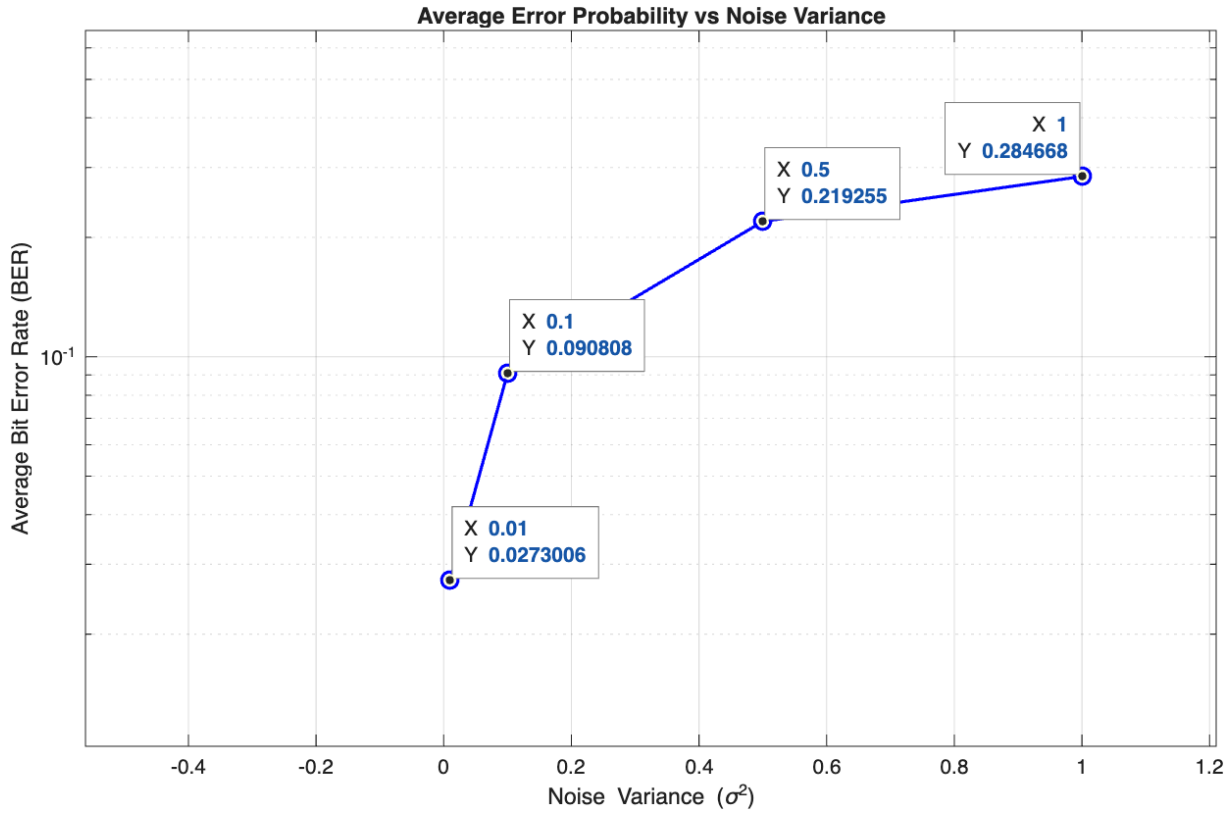


Figure 13: Noise variance (x axis) vs Average BER (y axis). Note that as variance increases, Average BER tends to fall and attain a stable slope.

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